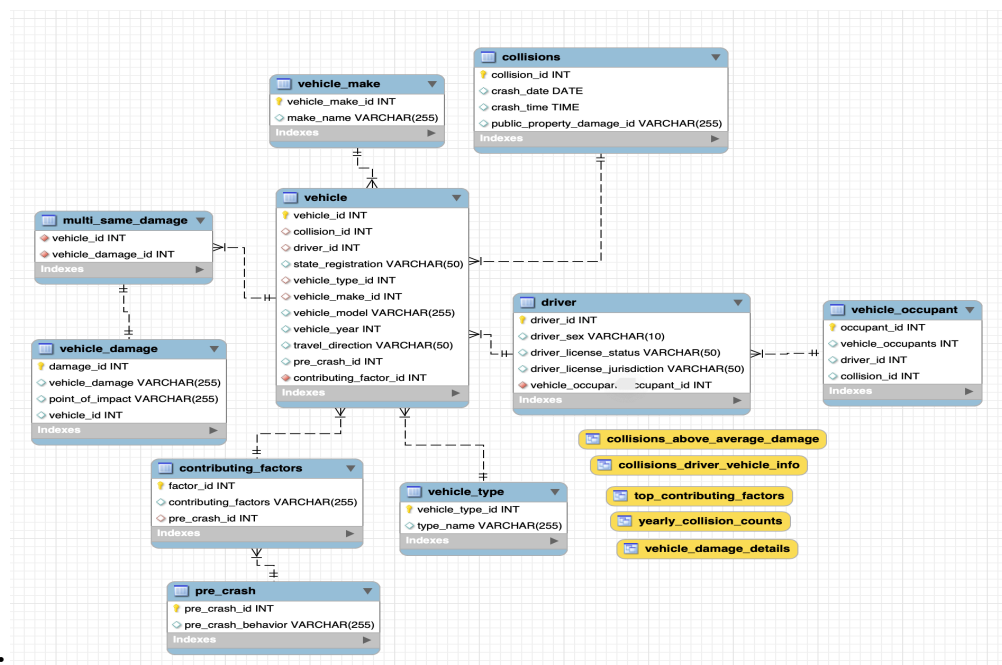


Final Project Report

Problem domain and motivation:

Due to the chaotic nature of the area, New York City has high statistics of motor vehicle collisions. Therefore, strategies should be implemented to improve traffic safety in highly active areas such as New York. Our project aims to construct a dataset that can be used to analyze traffic trends, identify high-risk areas, and incorporate other contributing factors to develop strategies to improve traffic safety. Through the use of this data, stakeholders can gain insights into the factors contributing to accidents and implement targeted interventions to reduce their occurrence. The chosen database will serve as a valuable resource for researchers, policymakers, and public safety officials aiming to enhance road safety in New York City, which will ultimately prevent fatalities of citizens.

Logical Design:



Our database is built to collect and organize detailed information about car crashes and what causes them. At the center of it is the Collision basically, each crash event where we record things like the date, time, and location. Since crashes usually involve more than one car, each collision can be linked to multiple Vehicles. Every vehicle is driven by one Driver, and we also track the Vehicle Type (like car, truck, or motorcycle) and Vehicle Make (the brand or manufacturer). To include more people involved, we have a part of the database for Vehicle Occupants, which covers passengers or anyone else in the vehicle during the crash. We also want to know why the crash happened, so we use Contributing Factors to keep track of causes like speeding or texting while driving. Each vehicle might also get damaged in the crash, so we use the Vehicle Damage section to describe what got damaged and how bad it was. Lastly, the Pre-Crash part helps explain what was going on just before the crash like if the car was turning, stopped, or speeding up. All of these pieces work together to show the full picture of what happened in a crash. This model helps everyone from police to researchers understand the crash better and design systems to prevent future ones. It also sets the stage for building more detailed parts of the database later.

Physical Design:

We built our database using MySQL, a relational database management system that organizes data in tables. Based on our ERD (Entity Relationship Diagram), we created separate tables for collisions, vehicles, and drivers. Each table has a primary key like `collision_id` or `vehicle_id` to keep records unique, and we used foreign keys to connect related data, for example, linking vehicles to collisions and drivers.

To make sure the tables connect properly, we added foreign key constraints, which stop incorrect data from being added, like a vehicle tied to a non-existent collision. We also used indexes to speed up searches, especially on fields like `crash_date`, and applied constraints like NOT NULL and UNIQUE to maintain data quality.

We had to be careful when loading data, entering records in the correct order to respect dependencies. Our database follows third normal form (3NF), which reduces data repetition by using lookup tables (like for vehicle type or make) and storing only their IDs in the main table. This keeps the data clean, easy to update, and ready for future growth.

Sample Data:

To populate our database tables, we used CSV files containing real or realistic sample data for each entity. We imported the data in a specific order to respect foreign key constraints, starting with lookup and parent tables (such as `vehicle_type`, `vehicle_make`, and `driver`) before importing dependent tables (such as `vehicle`, `collisions`, and `contributing_factors`). During the import process, we encountered challenges with NULL values, as some database systems do not interpret blank fields in CSVs as SQL NULL by default. To address this, we either replaced blank values with explicit NULL indicators in the CSV or handled them during the import process using import options or SQL scripts. This careful sequencing and data cleaning ensured referential integrity and successful data loading.

collision_id	crash_date	crash_time	public_property_damage_id	driver_driver_id
1	2012-09-07	9:03		1
2	2019-09-23	8:15	N	2
3	2015-10-02	17:18		3

driver_id	driver_sex	driver_license_status	driver_license_jurisdiction	vehicle_occupant_occupant_id
1				1
2	M	Licensed	NY	2
1				3

vehicle_id	collision_id	driver_id	state_registration	vehicle_type_id	vehicle_make_id	vehicle_model	vehicle_year	travel_direction	pre_crash_id
1	1	1	1 NY		1	1			1
2	2	2	2 NY		2	2	2002	North	2
3	3	3	1 NY		3	1			2

vehicle_make_id	make_name
1	NULL
2	TOYT -CAR/SUV
1	NULL

Views/Queries:

The **`collisions_driver_vehicle_info`** view combines data from the `collisions`, `driver`, `vehicle`, `vehicle_make`, and `vehicle_type` tables to display recent collisions (from 2020 onward) along with driver and vehicle details. This view uses multiple joins and filters the data by year, meeting the requirements for joins (A) and filtering (B).

The **`yearly_collision_counts`** view summarizes the number of collisions that occurred each year. It groups collision records by year and counts them, providing a clear trend over time. This view fulfills the requirements for aggregation (C) and the use of a join table (D).

The **top_contributing_factors** view identifies the most common contributing factors to collisions by joining the contributing_factors, vehicle, and pre_crash tables. It filters out null factors and aggregates the counts, showing the top ten factors along with related pre-crash behaviors. This view satisfies the requirements for joins (A), filtering (B), and aggregation (C).

The **vehicle_damage_details** view presents detailed information about vehicle damages by joining the multi_same_damage, vehicle, vehicle_damage, vehicle_make, and vehicle_type tables. It shows which vehicles suffered what kinds of damage and their associated make and type, fulfilling the requirements for joins (A) and the use of a join table (D).

The **collisions_above_average_damage** view highlights collisions that involved more vehicle damages than the average. It joins collisions, vehicle, and vehicle_damage tables, groups by collision, and uses a subquery to compare each collision's damage count to the overall average. This meets the requirements for joins (A), aggregation (C), and subqueries (E).

View name	Req. A (Join)	Req. B (Filter)	Req. C (Aggregation)	Req. D (Join Table)	Req. E (Subquery)
collisions_driver_vehicle_info	X	X			
yearly_collision_counts			X	X	
top_contributing_factors	X	X	X		
vehicle_damage_details	X			X	
collisions_above_average_damage	X		X		X

Changes from Original Design:

While working on our project, we made a lot of changes to our original database design. At first, our ERD was missing key tables, and many of the relationships didn't make sense or were based

on guesses rather than the actual data. After receiving feedback, we realized the design wouldn't work well for keeping data clean or supporting flexible queries. We went to office hours and got help from another TA, who explained how the relationships should work. With that guidance, we added missing tables, fixed foreign keys, and improved the overall structure based on the data's logic.

These changes helped us better normalize the database by removing duplicate data and keeping everything more organized. One unexpected challenge was importing the data. We ran into problems with NULL values in CSV files and had to load the tables in a specific order due to foreign key rules, which taught us how important planning is. In the end, our new design was much stronger. The data was cleaner, easier to manage, and allowed for better queries. The database did get more complex, which made some queries slower, but the benefits definitely outweighed the downsides. This process showed us how important it is to plan, test, and be flexible when building databases.

DEI:

Now that our database design is more developed, we can better evaluate its inclusiveness and address potential biases in how data is collected, structured, and interpreted. Our goal is to ensure the database provides fair insights into motor vehicle collisions in New York City without reinforcing inequalities or excluding vulnerable communities. While our dataset doesn't include direct demographic data like race or income, we recognize that marginalized groups are often more affected by traffic issues. Low-income neighborhoods, for example, may face poor road conditions or inadequate lighting. By using precise location-based data, we can highlight these patterns without needing sensitive personal information.

We also improved our filtering process to avoid leaving out areas with incomplete data. Instead of removing records from less-documented locations, we cleaned the data to retain key indicators like road conditions and accident frequency. This helps the database better reflect the city's geographic and socioeconomic diversity. By focusing on structural factors like road quality, time of day, and weather, rather than individual behavior, we shift the emphasis toward broader safety improvements. This promotes fairness and supports systemic change. We plan to keep reviewing

our design's inclusivity, gathering feedback, and staying aware of how the data might be misused. Our aim is to support all communities and encourage safer, more equitable streets.

Data Privacy:

We've taken extra steps to address privacy and ethical concerns in our project. The data we're using comes from the NYPD and is publicly available, with no personally identifiable information (PII). To be cautious, we're excluding or anonymizing any sensitive details, like specific addresses or combinations of time and location that might risk someone's privacy. We also reviewed the licensing rules for our sources, mainly from NYC Open Data, and confirmed the data is free to use under open licenses. We're following all terms—giving proper credit and avoiding any misuse or unauthorized changes—and we're staying transparent by citing our sources clearly.

Ethically, we recognize that even anonymized data can be misunderstood or misused without proper context. That's why we're including documentation in our final product to explain the data's limitations and highlight any potential biases, such as underreported areas or inconsistent reporting practices. To avoid reinforcing inequality, we made sure our data cleaning methods include a wide range of locations and incident types, not just the most documented ones. Overall, we've built our project on a foundation of responsible use, legal compliance, and transparency to ensure the database supports efforts to improve traffic safety effectively.

Potential Future Work:

To improve our database, we could connect it with other datasets like neighborhood demographics, transit routes, or weather data. This would add context and help reveal hidden trends in collision data. We could also expand time-based analysis by comparing accidents across seasons or days of the week, which might uncover useful safety patterns. Adding visual tools like charts or maps would make the data easier to understand without needing SQL, helping more people benefit from the insights. Finally, moving the project to a cloud platform like AWS or Google Cloud would support larger datasets, real-time updates, and API access for other tools. These upgrades would make the database more powerful, accessible, and useful in improving NYC traffic safety.

Lessons learned:

In our original ERD, we didn't have enough tables, and some relationships were incorrect. We believed it was accurate, but feedback revealed multiple errors. To fix it, we got help from a different TA during office hours. They helped us understand how to properly connect tables and identify the right fields for meaningful relationships.

We also had trouble importing CSV data—NULL values caused issues, so we had to clean the files. Another challenge was the import order: due to foreign key constraints, some tables had to be imported before others. This taught us the importance of planning and data dependencies. Through these challenges, we gained a stronger understanding of ERD design, data integrity, and relational database management.

Citations:

Kolker, Robert. "Why Traffic-Related Fatalities Are Rising after Years of Decline-- New York Magazine - Nymag." *New York Magazine*, New York Magazine, 24 Nov. 2012, nymag.com/news/features/traffic-2012-12/

"NYPD Motor Vehicle Collisions (Full Version): Nyc Open Data." *NYPD Motor Vehicle Collisions (Full Version) | NYC Open Data*, data.cityofnewyork.us/Public-Safety/NYPD-Motor-Vehicle-Collisions-Full-version-/3tta-b6xn Accessed 13 May 2025.

How NYC's Streets Became More Pedestrian-Friendly." *PBS NewsHour*, 11 June 2016, www.pbs.org/newshour/show/how-nycs-streets-became-more-pedestrian-friendly. Accessed 14 May 2025.

Titzer, Jon. "Why Traffic Accidents Remain a Leading Cause of Death in the U.S." *The Atlantic*, 21 Jan. 2023, www.theatlantic.com/health/archive/2023/01/traffic-accidents-leading-cause-death/673572/ Accessed 14 May 2025.